4 Chemical Equilibrium

SOLUTION (QR CODE)

5. First write the reaction:

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$,

Initial moles of N_2 = 4 mol

Initial moles of H_2 = 8 mol

Moles of NH_3 at equilibrium = 2 mol

	$1N_2(g)$	$+ 3H_2(g)$	$\rightleftharpoons 2NH_3(g)$
t = 0	4 mol	8 mol	0
$t = t_{eq}$	4 – x	8 – 3x	2x

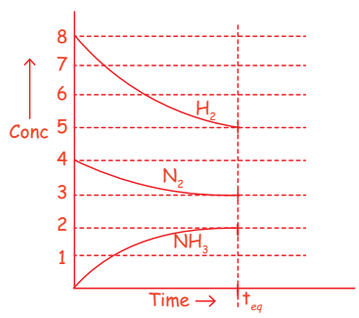
2x = 2, x = 1

Now, substitute the value of x to get $[N_2]$ and $[H_2]$ at equilibrium:

$[N_2]$ = 4 – 1 = 3 M

$[H_2]$ = 8 – 3(1) = 5 M

With time, the concentration of species will decrease.



graph tabhi hoga jab concentrations nikaloge

7. (182)

	A	+ B	⇌ C	+ D
Initial Conc.	1	1	1	1
At Equilibrium	1 – x	1 – x	1 + x	1 + x

$$K_{eq} = \frac{[C][D]}{[A][B]} = \frac{(1+x)(1+x)}{(1-x)(1-x)} = \left(\frac{1+x}{1-x}\right)^2$$

$$\left(\frac{1+x}{1-x}\right)^2 = 100 \rightarrow \frac{1+x}{1-x} = 10$$

∴ x = 0.818

[D] = 1 + x = 1 + 0.818 = 1.818

$[D] = 182 \times 10^{-2}$ M

11. (A) Total moles, n = 9 mol

Calculate P_{total} by using an ideal gas equation as:

$$P_{total} = \frac{nRT}{V} = \frac{(9 \times 0.082 \times 610)}{(100)} = 4.5\,atm$$

Now, calculate partial pressures of Ar and PCl_5 as:

$$P_{PCl_5} = \frac{5 \times 4.5}{9} = 2.5;\ P_{Ar} = \frac{4 \times 4.5}{9} = 2$$

PCl_5	⇌ PCl_3	+ Cl_2
2.5 – P	P	P

P_{total} = 2.5 – P + P + P + P_{Ar} = 6

P = 1.5

$$K_p = \frac{1.5 \times 1.5}{1} = 2.25$$

arey baccho ghabrao mat

13. [73]

	$AB_2(g)$	⇌ A(g)	+ 2B(g)
Initial	1 mole		
At equilibrium	1 – x mole	x mole	2x mole

At equilibrium, P = 1.9 atm

Total moles at equilibrium, n = 1 + 2x

Use ideal gas equation: PV = nRT

1.9 × 25 = (1 + 2x) × 0.08206 × 300

x = 0.465

Now; Partial pressure of AB_2 at equilibrium

$$= \frac{0.535}{1.93} \times 1.9\,atm$$

Partial pressure of A at equilibrium

$$= \frac{0.465}{1.93} \times 1.9\,atm$$

Partial pressure of B at equilibrium

$$= \frac{2 \times 0.465}{1.93} \times 1.9 \text{ atm}$$

Now; $K_p = \frac{P_A \cdot P_B^2}{P_{AB_2}}$

$$K_p = \frac{\left(\frac{0.465 \times 1.9}{1.93}\right)\left(\frac{2 \times 0.465}{1.93} \times 1.9\right)^2}{\frac{0.535}{1.93} \times 1.9}$$

$= 0.73 = 73 \times 10^{-2}$

17. (D) The relation between K_C and K_p is:

$K_p = K_C (RT)^{\Delta n_g}$

$K_p = K_C$ for those reactions in which the value of Δn_g is 0.

For reaction given in option (D), Δn_g is:

$\Delta n_g = 2 - 1 = 1$

So, for this reaction, K_p is not equal to K_C.

20. (C) The reaction is:

$SO_2Cl_{2(g)} \rightleftharpoons SO_{2(g)} + Cl_{2(g)}$

The relation between K_p and K_C is:

$K_P = K_C (RT)^{\Delta n_g}$, $\Delta n_g = 2 - 1 = 1$

Now, $\frac{K_p}{K_C} = (RT)^1$

As K_p is given in atm, R = 0.0821L atm/K/mol

Put $\frac{K_p}{K_C} = 3$, 3 = RT

$T = \frac{3}{0.0821}$, T = 36.54 K

direct formula hi lagaya hai or kuch nahi kiya

21. First write the reaction as:

$$A(g) + 2B(g) \longrightarrow 2C(g) + D(g)$$

Initial	x	1.5x		
Equilibrium	x - y	1.5x – 2y	2y	y

Now,

$K_p = K_C (RT)^{\Delta n} = K_C (RT)^{3-3} = K_C (RT)^0$

$= K_C \times 1 = K_C$

Also, $K_C = \frac{[C]^2[D]}{[B]^2[A]} = \frac{[2y \times 2y][y]}{[1.5x - 2y]^2[x - y]}$

Given equilibrium conc: x - y = 2y ...(1)

$K_p = \frac{2y^2}{[1.5x - 2y]^2}$ ...(2)

From equation (1), x = 3y

Put this value of x in equation (2), $K_p = 0.32$

x-2y ko agar 2y maan loge toh sawaal asaan ho jaega

22. (C,D) –1 here indicates the value of Δn_g.

Among the given reactions, reactions given in option (C) and (D) have –1 value of Δn_g.

28.

	$A_{2(g)}$	+ $B_{2(g)}$	$\rightleftharpoons$ $2AB_{(g)}$
$(t = t^{eq})_{old}$	2	2	2
$(t = t)_{disturb}$	2	2	1
$(t = t^{eq})_{new}$	2 - x	2 - x	1 + 2x

$$K_c = \frac{[AB]^2}{[A_2][B_2]} = \frac{\left(\frac{2}{V}\right)^2}{\left(\frac{2}{V}\right)\left(\frac{2}{V}\right)} = 1$$

$$K_c = \frac{\left(\frac{1+2x}{V}\right)^2}{\left(\frac{2-x}{V}\right)\left(\frac{2-x}{V}\right)} = 1$$

x = 1/3

$[A_2] = 2 - x = 2 - 1/3 = 1.67$ M

$[B_2] = 2 - x = 2 - 1/3 = 1.67$ M

$[AB] = 1 + 2x = 1 + 2(1/3) = 1.67$ M

30. Let a moles of NH_3 are added.

$$Ag^+(aq) + 2NH_3(aq) \rightleftharpoons Ag(NH_3)^+_2(aq)$$

At t = 0	0.8M	$\left(\frac{a}{2}\right)$M	———
At t = ∞	5×10^{-8}M	$\left(\frac{a}{2} - 1.6\right)$M	0.80 M

Here, K_f of complex ion $[Ag(NH_3)_2]^+$ is very high, so, we assume that all the Ag^+ is now converted into complex ion.

$$K_f = \frac{[Ag(NH_3)_2]^+}{[Ag^+][NH_3]^2}, \quad 10^8 = \frac{0.8}{(5 \times 10^{-8})\left(\frac{a}{2} - 1.6\right)^2}$$

$$\frac{0.8}{5} = \left(\frac{a}{2} - 1.6\right)^2, \left(\frac{a}{2} - 1.6\right) = 0.4$$

$$\frac{a}{2} = 0.4 + 1.6 \Rightarrow \frac{a}{2} = 2$$

Concentration of NH_3 added $= \frac{a}{2} = 2M$

Volume of solution = 2L

Moles of NH_3 added = 2 × 2 mol = 4 mol

31. (B) $HI \rightleftharpoons \frac{1}{2}H_2 + \frac{1}{2}I_2$; 8

$$\frac{1}{2}H_2 + \frac{1}{2}I_2 \rightleftharpoons HI ; \frac{1}{8}$$

$$2\left(\frac{1}{2}H_2\right) + 2\left(\frac{1}{2}I_2\right) \rightleftharpoons 2(HI) ; K'$$

Common multiplying factor = 2

$$K' = \left(\frac{1}{8}\right)^2 = \frac{1}{64}$$

34. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g), K_1 = 4 \times 10^5$....(i)

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g), K_2 = 1.6 \times 10^{12}$....(ii)

$H_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons H_2O(g), K_3 = 1.0 \times 10^{-13}$...(iii)

(ii) + 3 × (iii) – (i)

$$2NH_3(g) + \frac{5}{2}O_2(g) \rightleftharpoons 2NO(g) + 3H_2O(g)$$

$$k_{eq} = \frac{k_2 \times k_3^3}{k_1} = \frac{1.6 \times 10^{12} \times (10^{-13})^3}{4 \times 10^5}$$

$$= \frac{1.6}{4} \times 10^{-32} = 4 \times 10^{-33}$$

confuse kyu ho rahe ho?
kya baat ho gayi?

36. (A) The reaction is:

	$2AB_2(g)$	$\rightleftharpoons 2AB(g)$	$+ B_2(g)$
Initial moles	1	0	0
Moles at eq.	$1 - \alpha$	α	$\frac{\alpha}{2}$

Total moles at equilibrium = $1 - \alpha + \alpha + \frac{\alpha}{2} = \frac{2+\alpha}{2}$

$$K_p = \frac{(P_{AB})^2 (P_{B_2})}{(P_{AB_2})^2} = \frac{\left[\frac{2\alpha}{2+\alpha}P\right]^2 \left[\frac{\alpha}{2+\alpha}P\right]}{\left[\frac{2(1-\alpha)}{2+\alpha}P\right]}$$

$$= \frac{\alpha^3 P}{(2+\alpha)(1-\alpha)^2}$$

$$K_p \approx \frac{\alpha^3 P}{2} \text{ or } \alpha = \left[\frac{2K_p}{P}\right]^{\frac{1}{3}}$$

paglao mat bacho,
time do samjhne ke liye

38. (A) First write the dissociation of A_2B_3 as:

$A_2B_3 \rightleftharpoons 2A^+ + 3B^-$

Let c be the initial concentration of A_2B_3:

A_2B_3	$\rightleftharpoons 2A^+$	$+ 3B^-$
c	0	0
$c - c\alpha$	$2c\alpha$	$3c\alpha$

$$K_{eq} = \frac{(2c\alpha)^2 \times (3c\alpha)^3}{(c - c\alpha)} = \frac{108c^5\ \alpha^5}{(c - c\alpha)}$$

As α is small, $1 - \alpha$ can be approximated to 1

$K_{eq} \approx 108c^4\alpha^5$

On rearranging: $\alpha = \left(\frac{K_{eq}}{108c^4}\right)^{\frac{1}{5}}$.

A_2B_3 ki dissociation reaction
likh loge ya nahi?

41. [6] Moles of NH_4HS initially taken

$$= \frac{\text{Weight}}{\text{Molar Mass}} = \frac{5.1}{51} = 0.1 \text{ mol}$$

	$NH_4HS(s)$	$\rightleftharpoons NH_3(g)$	$+ H_2S(g)$
At t = 0	0.1 mol	0	0
At t = ∞	$0.1(1 - 0.2)$	0.1×0.2	0.1×0.2

According to ideal gas equation,

$$P_{NH_3} = \frac{nRT}{V} = \frac{0.1 \times 0.2 \times 0.082 \times 300}{2}$$

$$= 0.246 \text{ atm} = P_{H_2S}$$

$$K_p = P_{NH_3} \times P_{H_2S} = (0.246)^2$$

$$= 0.060516 = 6.05 \times 10^{-2} \approx 6 \times 10^{-2}$$

solid wale ko mat le lena expression me, ghonchu

45. (2 gm) $K_p = 1.2 = p_{CO_2} = \frac{n_{CO_2}.RT}{V}$

$n_{CO_2} = \frac{1.2 \times 2}{0.0821 \times 500} = 0.06$ mol

Decomposed moles of $CaCO_3$ = 0.06 mol

$n_{CaCO_3} = 0.06$

$W_{CaCO_3} = 0.06 \times 100g = 6$ gm

Decomposed moles of $CaCO_3$ = 0.06 mol = 6 gm

Left $CaCO_3$ = 8 gm – 6 gm = 2 gm

46. (6 atm) $NH_2COONH_{4(s)} \rightleftharpoons 2NH_{3(g)} + CO_{2(g)}$

2P P

$K_p = P^2_{NH_3}.P_{CO_2} = (2P)^2 (P)$

$32 = 4P^3$, P = 3 atm

$P_{total} = P_{NH_3} + P_{CO_2} = 2P + P = 3P$ atm

$P_{total} = 3 \times 2 = 6$ atm

49. (0.833 atm) First calculate the moles of $CaCO_3$ as:

$$\text{Moles} = \frac{\text{Mass}}{\text{molar mass}} = \frac{20}{(100)} = \frac{1}{5}\text{atm}$$

It is given that 75% of $CaCO_3$ is decomposed, so,

$$75\% \text{ of } \frac{1}{5} = \frac{3}{20}\text{mol}$$

$CaCO_{3(s)}$	$\rightleftharpoons$	$CaO_{(s)}$	+	$CO_{2(g)}$
$\frac{1}{5}$mol		0		0
$\left(\frac{1}{5} - \frac{3}{20}\right)$mol		$\frac{3}{20}$mol		$\frac{3}{20}$mol

$K_p = P_{CO_2}$

Calculate P using an ideal gas equation:

$$P = \frac{xRT}{V} = \frac{\frac{3}{20} \times \frac{1}{12} \times 1000}{15}\text{atm} = 0.833 \text{ atm}$$

50. $(P_{total})_{old} = 3P$

$K_p = P^2_{NH_3} . P_{CO_2} = (2P)^2 P = 4P^3$ (Initial Eq^m)

At New eq^{bm}, $P'_{NH_3} = 3P$

$K_p = 4P^3 = P'^2_{NH_3} . P'_{CO_2}$

$4P^3 = (3P)^2 P'_{CO_2}$

$P'_{CO_2} = \frac{4P}{9}$

$(P_{Total})_{new} = P'_{NH_3} + P'_{CO_2} = 3P + \frac{4P}{9} = \frac{31P}{9}$

$$\frac{(P_{Total})_{new}}{(P_{Total})_{old}} = \frac{\frac{31P}{9}}{3P} = \frac{31}{27}$$

hai na mazedaar?

56. (B) First write both reactions as:

$A_{(s)} \rightleftharpoons B_{(g)} + C_{(g)}$

a a + b

$D_{(s)} \rightleftharpoons C_{(g)} + E_{(g)}$

a + b b

$K_{P_1} = x = a (a + b)$ ...(1)

$K_{P_1} = b = b (a + b)$ ...(2)

On adding (1) and (2)

$(a + b)^2 = x + y$

$a + b = (x + y)^{\frac{1}{2}}$

Now, $P_{total} = 2(x + y)^{\frac{1}{2}}$

67. (D) In given equation,

$CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$

If we add inert gas there will be no effect as volume is constant

At constant pressure, for the reactions for which $\Delta n_g = 0$, there is no effect of addition of inert gas to a reaction in equilibrium.

The presence of a catalyst does not effect the amount of a reactant or a product at equilibrium.

When the reactant CO concentration is increased, the equilibrium shifts forward and the product concentration is increased.

Kuch hui halchal dimag mei ya nahi?

76. (74)

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$$

At t = 0 0.1 mol a mol 0

At t = eq 0.1 – x a – 2x x

According to the question, x = 0.04

$\therefore$ $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$

At t = 0 0.1 a 0

At t = eq 0.1 – 0.04 a – (2×0.04) x = 0.04 mole

= 0.06 mole = –0.08 mole

Given: V = 2L, T = 500 K, P_{total} = 5 bar

n_{total} = 0.25

$$P_{total} = n_{total} \times \frac{RT}{V}$$

$$\Rightarrow 5 = (0.06 + (a - 0.08) + 0.04) \times \frac{0.08 \times 500}{2}$$

$$\Rightarrow 5 = (0.02 + a) \times \frac{0.08 \times 500}{2}$$

$\Rightarrow$ a = 0.23 mol

Now,

$$K_p = \frac{X_{CH_3OH}}{X_{CO} \times X_{H_2}^2} \times \frac{1}{(P_T)^2}$$

$$= \frac{0.04}{0.06 \times (0.15)^2} \times \left[\frac{1/4}{5}\right]^2$$

$$= \frac{4}{6 \times 0.15 + 0.15} \times \frac{1}{16 \times 25}$$

$$= \frac{4 \times 100 \times 100}{6 \times 15 \times 15 \times 16 \times 25}$$

$= 0.074 = 74 \times 10^{-3}$

77. (B)

$$A(g) \rightleftharpoons B(g) + C(g)$$

At t = 0 a 0 0

At t = 1 a(1 – α) aα aα

Now, mole fraction of A:

ΞA = α – αα / α – αα + αα + αα

$$= \frac{a - a\alpha}{a + a\alpha} = \frac{1 - \alpha}{1 + \alpha}$$

Mole fraction of B:

$$X_B = \frac{a\alpha}{a + a\alpha} = \frac{\alpha}{1 + \alpha}$$

Mole fraction of C:

$$X_C = \frac{a\alpha}{a + a\alpha} = \frac{\alpha}{1 + \alpha}$$

Let's assume if P is the total pressure, then partial pressure of A(g), B(g) and C(g) are as follows:

$$P_A = \left(\frac{1-\alpha}{1+\alpha}\right)P,\ P_B = \left(\frac{\alpha}{1+\alpha}\right)P \text{ and } P_C = \left(\frac{\alpha}{1+\alpha}\right)P$$

Therefore,

$$K_p = \frac{\left(\frac{\alpha}{1+\alpha}\right)P\left(\frac{\alpha}{1+\alpha}\right)P}{\left(\frac{1-\alpha}{1+\alpha}\right)P} \Rightarrow K_p = \frac{\left(\frac{\alpha}{1+\alpha}\right)^2 P^2}{\left(\frac{1-\alpha}{1+\alpha}\right)P}$$

$$K_p = \left(\frac{\alpha}{1+\alpha}\right) \times \left(\frac{\alpha}{(1+\alpha)}\right) \times \left(\frac{1+\alpha}{1-\alpha}\right).P$$

$$K_p = \frac{\alpha^2}{(1+\alpha)(1-\alpha)}.P$$

$$K_p = \frac{\alpha^2 P}{1-\alpha^2}$$

Since, K_p is only function of temperature.

Therefore, as P increase, α decreases.